

Project Size Estimation Report

Project Name: KuCampusBite

Project Overview: This is a web-based platform for the University of Karachi that enables students to order food from campus canteens. The system displays menus, provides fast delivery, and sends order details directly to canteens via WhatsApp, making food more accessible and saving students' time.

Users: Canteen staff , system admin , all users(students ,teachers)

Project Scope: User login & registration,menu browsing& search,food ordering , food reviewing & scheduling , admin dashboard & Whatsapp notification to canteens.

Functional Point	Type	Complexity
Login Form	EI	simple
Signup Form	EI	Average
Add to Cart	EI	simple
Place Order Form	EI	Average
Review Submission	EI	simple
Schedule Order	EI	High
Admin Login	EI	simple
Admin Add Canteen	EI	Average
Admin Update Canteen/Menu	EI	High
Order Confirmation Page	EO	Average
WhatsApp Notification	EO	Average
Total Sales Report	EO	Average
Food/Canteen Search	EQ	Average
Canteen Menu Page	EQ	Average
Overall Menu Page	EQ	Average
Food Item Detail Page	EQ	Average
My Cart Page	EQ	Average
Order Placed Page	EQ	Average
Review Display	EQ	Simple
User Data	ILF	Average
Canteen Data	ILF	Average
Food Menu Data	ILF	High
Review Data	ILF	Simple
Cart Data	ILF	Average
Order Data	ILF	High
WhatsApp API	EIF	Average

Unadjusted Function Points Calculation:

$$EI = 4 \times 3 + 3 \times 4 + 2 \times 6 = 36$$

$$EO = 3 \times 5 = 15$$

$$EQ = 1 \times 3 + 6 \times 4 = 27$$

$$ILF = 1 \times 7 + 3 \times 10 + 2 \times 15 = 67$$

$$EIF = 1 \times 7 = 7$$

$$\mathbf{UFP} = \Sigma(\text{Count} \times \text{Complexity Multiplies}).$$

$$\mathbf{UFP} = 36 + 15 + 27 + 67 + 7 = 152.$$

Total GSC (TDI):

$$\begin{aligned} &= 4 + 1 + 4 + 3 + 3 + 2 + 4 + 3 + 2 + 3 + 3 + 2 + 3 + 3 \\ &= 40 \end{aligned}$$

GSC #	Characteristic	Rating
1	Data Communications	4
2	Distributed Data Processing	1
3	Performance	4
4	Heavily Used Configuration	3
5	Transaction Rate	3
6	Online Data Entry	2
7	End-User Efficiency	4
8	Online Update	3
9	Complex Processing	2
10	Reusability	3
11	Installation Ease	3
12	Operational Ease	2
13	Multiple Sites	3
14	Facilitate Change	3

CAF:

$$\begin{aligned} &= 0.65 (0.01 \times 40) \\ &= 1.05 \end{aligned}$$

AFP:

$$\begin{aligned} &= \text{UFP} \times 1.05 \\ &= 152 \times 1.05 \\ &= 159.6 \end{aligned}$$

Based on the Function Point Analysis (FPA) method, the estimated project size for KuCampusBite is 159.6 Adjusted Functional Points (AFP).